Ki-won Eom 엄기원

Undergraduate researcher, AIM Lab, Hanbat National University
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Research interests

Computer vision, object detection in satellite imagery (optical, aerial and SAR), image forensics

Education

B.S. in Computer Engineering, Hanbat National University, Daejeon, Korea · Mar 2023 - present

Research experience

Undergraduate researcher, AIM Lab, Hanbat National University · Advisor: Prof. Haneol Jang · Feb 2024 - present

- Research project: Sub-project 5 (intelligent processing of puzzle-type ultra-wide-area satellite imagery) of the Space-layer Intelligent Communication Network Specialized Research Lab (KRIT).
- Designed and implemented SPOT-DETR, a detector for tiny objects in aerial images (co-first author). It reaches 37.6 AP on AI-TOD and 33.3 AP on AI-TOD-v2, above every detector retrained under the same protocol.
- Proposed YOLO-Dos, a speckle suppression module for SAR ship detection (first author). It raises mAP@50:95 by 5.75 points over YOLOv8 on HRSID.
- Proposed region weights for the eyes, nose and mouth in a ViT deepfake detector (first author). On FaceForensics++ it improves ACER by 7.94% and AUC by about 3% over plain ViT.

Publications

† equal contribution

Ki-won Eom, Eunkyung Kim, Haneol Jang. **YOLO-Dos: Improving YOLO-based SAR object detection performance via a speckle suppression module.** Journal of IKEEE, 30(1):171-181, 2026 · KCI-indexed (in Korean). DOI

Ki-won Eom, Dong-Su Kim, Tae-Woo Oh, Haneol Jang. **ViT-based deepfake detection via regional emphasis weights on eyes, nose, and mouth.** Journal of Digital Forensics, 19(4):116-131, 2025 · KCI-indexed (in Korean). DOI

Ki-won Eom†, Cheoneum Park†, Eunkyung Kim, Haneol Jang. **SPOT-DETR: Tiny object detection in aerial images using high-resolution feature refinement and off-grid candidate memory.** Manuscript in preparation, 2026. project page

Honors and competitions

Grand Prize, 6th AI SPARK Challenge · wildfire detection from multispectral satellite imagery · 2024

Finalist (96 of 3,393) · 18th of 33 teams in the final, LG Aimers 9th cohort hackathon · 29th of 1,090 teams online (top 3%) · 2026

6th of 138 teams (top 5%), National Park Satellite Monitoring AI Challenge · coastal debris · 2026

16th of 271 teams (top 6%), SW-Centered University Digital Competition · 2025

85th of 985 teams (top 9%), Wind Power Forecasting AI Competition (BARAM 2026) · 2026

28th of 219 teams (top 13%), SW-Centered University Digital Competition · 2024

2nd place, cloud segmentation Kaggle challenge · Hanbat National University AI course · 2025

Projects

SPOT-DETR web demo (capstone design project) · 2026

- Web demo comparing SPOT-DETR with the baseline side by side; large images are tiled at native resolution.

AI-based detection of unpermitted livestock barns (with K-water) · 17,126 aerial images; YOLO11 ensemble, 90.7% precision, 82.9 mAP@0.5 · 2026 - in progress

Cloud segmentation in satellite imagery (team lead) · 2nd place in a course Kaggle challenge (0.8555) · Sep - Dec 2025

Face anti-spoofing classification (team lead) · 98.33% accuracy on live vs. spoofed faces · Mar - Jun 2025

Face-recognition smart door lock (team lead) · Raspberry Pi 4 · OpenCV · SQLite · MQTT · Sep - Dec 2024

Skills

Languages Python, C, Java
Deep learning PyTorch, MMDetection, Ultralytics YOLO, timm, OpenCV, MediaPipe, scikit-learn
Infrastructure Linux multi-GPU servers, Docker, Git, Weights & Biases